CSDS 412 Group 4 Project Status

Abstract

Parkinson's disease (PD) is a chronic, heterogeneous neurodegenerative disorder for which clinicians lack objective biomarkers to reliably track disease progression and forecast symptom severity over time. This project investigates the use of data science and parallel programming techniques to predict PD progression from longitudinal molecular biomarkers. Leveraging the Kaggle AMP-Parkinson's Disease Progression Prediction dataset, we will model repeated cerebrospinal fluid protein/peptide measurements and visit timing to forecast clinical severity, specifically MDS-UPDRS symptom scores. Our goal is to enable earlier risk stratification and better targeting for care and clinical trials.

We adopt a Random Forest model as a robust, non-linear baseline. Our methodology focuses on developing an end-to-end, reproducible pipeline for high-dimensional longitudinal data, encompassing data ingestion, validation, and efficient feature construction (e.g., protein/peptide aggregation, longitudinal deltas, and missingness indicators). The evaluation will employ patient-level splits and time-aware validation. To support rapid iteration and portability, we will parallelize preprocessing and training using OpenMP, track experiments and artifacts, and containerize the workflow with Singularity for execution on shared compute environments. We will compare regularized linear baselines with tree-based ensembles (Random Forest and gradient boosting). We hypothesize that incorporating changes in biomarkers over time will improve predictions of future PD symptom severity, with tree-based models outperforming simpler linear approaches. Furthermore, the integration of parallel computing and containerized workflows is expected to enhance the analysis's speed, reliability, and reproducibility.

Introduction

Parkinson’s disease (PD) is a chronic and progressive neurodegenerative disorder characterized by motor symptoms such as tremor, rigidity, and bradykinesia, as well as non-motor symptoms including cognitive impairment and sleep disturbances (Yamashita et al., 2023). A key challenge in PD management is its heterogeneity: patients exhibit widely varying rates and patterns of symptom progression over time (Holden et al., 2017). Because disease trajectories differ significantly across individuals, clinicians often lack objective tools that reliably forecast future symptom severity. This limitation affects early risk stratification, personalized treatment planning, and clinical trial targeting.

Advances in high-throughput molecular profiling now allow large-scale measurement of cerebrospinal fluid (CSF) proteins and peptides that may serve as biomarkers of disease progression (Yamashita et al., 2023). Longitudinal datasets containing repeated biomarker measurements aligned with clinical visits create an opportunity to model temporal changes in disease state rather than relying solely on baseline information. Clinical severity in PD is commonly quantified using the Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS), which tracks symptom burden over time (Holden et al., 2017). Predicting future MDS-UPDRS scores from longitudinal molecular biomarkers could therefore improve understanding of disease trajectories and provide a data-driven framework for forecasting progression.

The central research question of this project is: ***Can longitudinal CSF protein and peptide measurements, combined with visit time information, improve prediction of future MDS-UPDRS scores compared to regularized linear models?*** To address this question, we leverage the AMP-Parkinson's Disease Progression Prediction dataset from Kaggle, which contains repeated CSF protein and peptide measurements aligned with clinical visit timing. We develop a reproducible, end-to-end data science pipeline that performs data ingestion and validation, feature engineering (including protein aggregation and longitudinal delta and trend computation), and evaluation using patient-level splits and time-aware validation strategies.

From a modeling perspective, we implement Random Forest as a non-linear baseline and compare its performance with regularized linear models (Ridge and Lasso) and gradient boosting approaches. From a systems perspective, we incorporate parallel preprocessing and model training strategies to improve runtime efficiency for high-dimensional longitudinal data. The entire workflow is containerized using Singularity to ensure portability and reproducibility across shared compute environments (Sochat et al., 2017).

We hypothesize that incorporating longitudinal biomarker trends will improve the prediction accuracy over static features and that tree-based ensemble methods will outperform linear baselines while maintaining scalable and reproducible execution.

Accomplishments

- **Initial Setup:**
    - Created a shared GitHub repository for team collaboration.
    - Migrated datasets to the group's shared space.
    - Cleaned and organized the data, preparing it for immediate modeling.
- **Project Objective:** We clearly defined the goal for our project: To predict future MDS-UPDRS scores using longitudinal CSF protein and peptide biomarkers. Comparison: Evaluate and track both the accuracy and runtime of linear models versus tree-based methods (e.g., XGBoost).
- **Data Acquisition and Early Analysis:** We acquired the AMP-PD Progression dataset from Kaggle. We have completed initial profiling, assessing dimensionality, missingness, visit frequency, and score distributions. Some of the core challenges we noticed: High dimensionality and longitudinal sparsity.
- **Tech Stack Selection:** We have identified the technologies that we will use in this project. Data Processing: Python (Pandas, NumPy). Modeling: scikit-learn and XGBoost. Performance: Parallel preprocessing and runtime benchmarking on HPC. Reproducibility: Singularity for consistent runs on shared compute resources.

- **Initial Data Preprocessing Pipeline:** We built the first initial version of the pipeline: schema checks, missingness handling, basic normalization, protein/peptide aggregation, and construction of longitudinal features (deltas and simple trends). Generated model training and validation splits at the patient level.

Methodology

1. Data Pipeline

- Schema validation
- Outlier detection and normalization
- Protein/peptide aggregation
- Longitudinal delta and trends computation

2. Modeling

- Linear Regression and Ridge/Lasso
- Random Forest and XGBoost (Baseline)
- Gradient Boosting

3. Evaluation

- SMAPE, RMSE, MAE, R^2
- Patient-level splits
- Time-aware validation
- Runtime comparison, both serial and parallel

Work Remaining

- **Feature Engineering Refinement:** Expand longitudinal features (rolling windows, slope estimation, lag features), improve handling of sparsity, and evaluate alternative aggregation strategies at protein vs peptide level.
- **Model Development & Tuning:** Complete implementation of all baselines (Ridge, Lasso, Random Forest, XGBoost), perform systematic hyperparameter tuning, and compare performance across validation schemes.
- **Parallelization Optimization:** Benchmark serial vs parallel preprocessing and training; optimize OpenMP-backed components and evaluate scalability with increasing data size.
- **Evaluation, Results Analysis + Reporting:** Run experiments and evaluation, conduct feature importance analysis, interpret model behavior, compare runtime-performance tradeoffs, and finalize report and visualizations.

Responsibilities

Task	Assigned Members

Data acquisition, validation, cleaning	Nat Wang , Lakshmi Poojitha Annapureddy
Feature engineering: delta, trends, aggregation logic	Nat Wang , Lakshmi Poojitha Annapureddy
Baseline models implementation	Aiden Le , Ha Phan
Gradient boosting, feature importance analysis	Khue Luong , Khoa Luong
Parallel preprocessing, runtime benchmarking, cluster scripts	Aiden Le , Ha Phan
Singularity container, experiment tracking, report production	All members

References

https://www.kaggle.com/c/amp-parkinsons-disease-progression-prediction/overview
https://www.sciencedirect.com/science/article/pii/S266614462300014X (Yamashita et al., 2023)
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0188511 (Sochat et al., 2017)
https://pubmed.ncbi.nlm.nih.gov/29662921/ (Holden et al., 2017)